Ans : (a) Lion-tailed macaque is found in the Western Ghats of India. It is an endangered species, which is found mostly in Kerala, Karnataka and the border regions of Tamil Nadu. It is not found in Andhra Pradesh.

94. Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms ?

(a) Ecotone (b) Ecological niche
(c) Habitat (d) Home range

Ans : (b) In nature, many species occupy the same habitat, but they perform different functions. The functional characteristics of a species in its habitat are referred to as 'niche' in that common habitat. Ecological niche refers to both the ecological role and the space that an organism occupies in the community of organisms.

95. Photochemical smog is a resultant of the reaction among –

(a) NO_2, O_3 and peroxyacetyl nitrate in the presence of sunlight
(b) CO, O_2 and peroxyacetyl nitrate in the presence of sunlight
(c) CO, CO_2 and NO_2 at low temperature
(d) high concentration of NO_2 O_3 and CO in the evening

Ans : (a) Major chemical pollutant in photochemical smog :
(i) Nitrogen oxides (NO and NO_2)
(ii) Volatile Organic Compounds (VOC_5)
(iii) Ozone (O_3)
(iv) Peroxyacetyl Nitrates (PAN)

96. Consider the following :

1. Star tortoise 2. Monitor lizard
3. Pygmy hog 4. Spider monkey

Which of the above are naturally found in India?

(a) 1, 2 and 3 only (b) 2 and 3 only
(c) 1 and 4 only (d) 1, 2, 3 and 4

Ans : (a) Spider monkeys live in tropical climates, specifically the evergreen forests of Central and South America. They are not found in India. Star tortoise is found in India in the dry and scrab forests. Pygmy Hog is an endangered species found in Assam. Monitor Lizard is found in India, Sri Lanka and Pakistan.

97. Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity ?

(a) Oceans, lakes, grasslands mangroves
(b) Mangroves, oceans, grasslands, lakes
(c) Mangroves, grasslands, lakes, oceans
(d) Oceans, mangroves, lakes, grasslands

Ans : (c) In order of decreasing productivity of an ecosystem, their correct sequence is – Mangrove > grassland > lakes > oceans.

98. Which of the following can be found as pollutants in the drinking water in some parts of India ?

1. Arsenic 2. Sorbitol
3. Fluoride 4. Formaldehyde
5. Uranium

Select the correct answer using the codes given below:

(a) 1 and 3 only (b) 2,4 and 5 only
(c) 1,3 and 5 (d) 1, 2, 3, 4 and 5

Ans : (c) Uranium has been found in the water of Punjab and Fluoride in the water of Andhra Pradesh, Gujarat, Haryana, Odisha, Punjab etc. Arsenic has been found in Haryana and West Bengal, among others. Sorbitol, on the other hand, is a bulk sweetener found in food products. Formaldehyde is an organic compound, which is toxic to human beings. It is a gas at room temperature. It is not a water pollutant in India.

99. Contour bunding is a method of soil conservation used in –

(a) desert margins, liable to strong wind action
(b) low flat plains, close to stream courses, liable to flooding
(c) scrublands, liable to spread of weed growth
(d) None of the above

Ans : (d) Contour bunding (or contour bundling), involves the placement of lines of stones along with the natural rises of a landscape. It helps to capture and hold rainfall before it can become runoff which reduces soil erosion and retards overall water loss.

MISCELLANEOUS

100. In the context of cultural history of India, a pose in dance and dramatics called Tribhanga' has been a favorite of Indian artists from ancient times till today. Which one of the following statements best describes this pose ?

(a) One leg is bent and the body is slightly but oppositely curved at waist and neck
(b) Facial expressions, hand gestures and make up are combined to symbolize certain epic or historic characters
(c) Movements of body, face and hands are used to express oneself or to tell a story
(d) A little smile, slightly curved waist and certain hand gestures are emphasized to express the feelings of love or eroticism

Ans : (a) Tribhanga literally meaning three parts break, consists of three bends in the body, at the neck, waist and knee; hence the body is oppositely curved at waist and neck, which gives it a gentle 'S' shape.

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2012

GENERAL STUDIES : PAPER-I

Time : 2 hours **(Exam date : 20.05.2012)** **Maximum Number : 200**

ANCIENT HISTORY

• Pre-Historic Age

1. With reference to the scientific progress of ancient India, which of the statements given below are correct?

1. Different kinds of specialized surgical instruments were in common use by 1st century AD.
2. Transplant of Internal organs in the human body had begun by the beginning of 3rd century AD.
3. The concept of sine of angle was known in 5th century AD.
4. The concept of cyclic quadrilaterals was known in 7th century AD.

Select the correct answer using the codes given below :

(a) 1 and 2 only (b) 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

Ans. (c): There was significant progress in India in ancient times. Various sources suggest that by the first century AD, a variety of surgical instruments were in use, but organ transplantation was not possible until the third century AD. By the 5th century AD, in the area of mathematics in the Gupta period, the principles of sine, cosine and inverse sine were known in India. They are mentioned in 'Surya Siddhanta' and 'Aryabhatiya'. In the seventh century CE, the theory of the cyclic quadrilateral was known by Brahmagupta.

• Vedic Civilization

2. The religion of early Vedic Aryans was primarily of :

(a) Bhakti.
(b) Image worship and Yajnas.
(c) Worship of nature and Yajnas.
(d) Worship of nature and Bhakti.

Ans. (c): Vedic deities were predominantly Indra, Agni (fire), Soma, Mitra, Varuna, Ushas, Aditi & Rudra. Aryans used to worship nature deities such as Surya (Sun), Vayu (Wind) and Prithvi (Earth). Rivers, especially Saraswati, were worshipped as the goddess. Yajna was the mode of worship, which was the performance of sacrifices. It included vedic chants, samans singing and sacrificial mantra i.e, yajus. Yajna involved sacrificing havan samagri in the fire while chanting the vedic mantra.

• Buddhism, Jainism, Bhagavata, Shaiva and Other Religion

3. Lord Buddha's image is sometimes shown with the hand gesture called 'Bhumisparsha Mudra' It symbolizes.

(a) Buddha's calling of the Earth to watch over Mara and to prevent Mara from disturbing his meditation.
(b) Buddha's calling of the Earth to witness his purity and chastity despite the temptation of Mara.
(c) Buddha's reminder to his followers that they all arise from the Earth and finally dissolve into the Earth, and thus this life is transitory.
(d) Both the statements (a) and (b) are correct in this context.

Ans. (b): The 'Bhumisparsha Mudra' is one of the most common iconic images of Buddhism. It depicts the Buddha sitting in meditation with his left hand, palm upright, in his lap, and his right hand touching the earth. This represents the moment of the Buddha's enlightenment.

4. With reference to the history of ancient India, which of the following was/were common to both Buddhism and Jainism?

1. Avoidance of extremities of penance and enjoyment.
2. Indifference to the authority of the Vedas.
3. Denial of efficacy of rituals.

Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3 only
(c) 1and 3 only (d) 1, 2 and 3

Ans. (b): The avoidance of excess asceticism and indulgence and following the middle path is found only in Buddhism. While ideologies such as disobedience to the authenticity of the Vedas, prohibitions of the virtue of rituals and prohibitions of violence of beings, etc., are found in both Buddhism and Jaina religions alike.

MEDIEVAL HISTORY

• Vijayanagara Empire and Bahmanids

5. The Nagara, the Dravida and the Vesara are the

(a) three main racial group of the Indian subcontinent.
(b) three main linguistic division into which the languages of India can be classified.
(c) three main styles of Indian temple architecture.
(d) three main musical Gharanas prevalent in India.